

1 **Supplementary Materials for**

2
3 Detecting shifts in nonlinear dynamics using Empirical Dynamic Modeling with Nested-Library
4 Analysis

5
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11 **This supplement file includes:**
12 **S1 Text**
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S1 Text Pseudocodes of Nested-Library Analysis

Algorithm 1: Nested-Library Analysis, checking the left part of the time series

Data: Time series $\{x_t\}_{t=0}^L$ as the library set, $\{x_t\}_{t=L+1}^R$ as the test set, and a number $D_{skipstep} \in \mathbb{N}$

Result: An estimated time $\hat{t} \in [0, L)$ of a regime shift if it exists

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for ( $n = 0; n < L; n = n + D_{skipstep}$ ) do
  /* Step 1. Monitor the prediction error curve  $\tilde{\mathcal{E}}_n$  */
   $\theta_n \leftarrow \arg \min_{\theta} \{\sum_{t=L+1}^R (y_t - x_t)^2 \mid y_t: \text{the prediction of } x_t \text{ given by S-map } (\{x_t\}_{t=n}^L; \theta)\};$ 
  // the optimal parameter for S-map to predict  $\{x_t\}_{t=L+1}^R$  with  $\{x_t\}_{t=n}^L$ .
  for ( $l = n; l < L; l = l + D_{skipstep}$ ) do
     $\{y_t^{(l)}\}_{t=L+1}^R \leftarrow \text{the prediction of } \{x_t\}_{t=L+1}^R \text{ given by S-map } (\{x_t\}_{t=l}^L; \theta_n);$ 
     $\mathcal{E}_n(l) \leftarrow \sqrt{\sum_{t=L+1}^R \frac{(x_t - y_t^{(l)})^2}{R-L}};$ 
  end
  /* Step 2. Find the change point  $\tau_n$  */
   $\tilde{\mathcal{E}}_n \leftarrow \text{the smoothened } \mathcal{E}_n \text{ by Gaussian filter};$ 
  if  $\tilde{\mathcal{E}}_n$  is valley-shaped then
     $\hat{t}_n \leftarrow \arg \min_t \tilde{\mathcal{E}}_n(t);$ 
  else
     $\hat{t}_n \leftarrow \text{NaN};$ 
  end
end
/* Step 3. Vote for the output */
 $\hat{t} \leftarrow \text{median of } \{\hat{t}_n\}_n;$ 
return  $\hat{t};$ 

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Algorithm 2: Nested-Library Analysis, checking the right part of the time series

Data: Time series $\{x_t\}_{t=L+1}^R$ as the library set, $\{x_t\}_{t=0}^L$ as the test set, and a number $D_{skipstep} \in \mathbb{N}$

Result: An estimated time $\hat{t} \in (L + 1, R]$ of a regime shift if it exists

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for ( $n = R; n > L + 1; n = n - D_{skipstep}$ ) do
  /* Step 1. Monitor the prediction error curve  $\tilde{\mathcal{E}}_n$  */
   $\theta_n \leftarrow \arg \min_{\theta} \{\sum_{t=0}^L (y_t - x_t)^2 \mid y_t: \text{the prediction of } x_t \text{ given by S-map } (\{x_t\}_{t=L+1}^n; \theta)\};$ 
  // the optimal parameter for S-map to reconstruct  $\{x_t\}_{t=0}^L$  with  $\{x_t\}_{t=L+1}^n$ .
  for ( $l = n; l > L + 1; l = l - D_{skipstep}$ ) do
     $\{y_t^{(l)}\}_{t=0}^L \leftarrow \text{the prediction of } \{x_t\}_{t=0}^L \text{ given by S-map } (\{x_t\}_{t=l}^n; \theta_n);$ 
     $\mathcal{E}_n(l) \leftarrow \sqrt{\sum_{t=0}^L \frac{(x_t - y_t^{(l)})^2}{L}};$ 
  end
  /* Step 2. Find the change point  $\tau_n$  */
   $\tilde{\mathcal{E}}_n \leftarrow \text{the smoothened } \mathcal{E}_n \text{ by Gaussian filter};$ 
  if  $\tilde{\mathcal{E}}_n$  is valley-shaped then
     $\hat{t}_n \leftarrow \arg \min_t \tilde{\mathcal{E}}_n(t);$ 
  else
     $\hat{t}_n \leftarrow \text{NaN};$ 
  end
end
/* Step 3. Vote for the output */
 $\hat{t} \leftarrow \text{median of } \{\hat{t}_n\}_n;$ 
return  $\hat{t};$ 

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Remark. The valley-shapedness of curves is determined by the discriminant in Sec. 2.2.

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S2 Text Sequential locally weighted global linear map (S-map)

For a chosen dimension m , a delay term τ (which does not stand for the dynamical change point here), and a given time series $\{x_t\}_t$, the attractor is reconstructed as $\{X_t\}_t$ with

$$X_t := (1, x_t, x_{t-\tau}, \dots, x_{t-(m-1)\tau}) \in \mathbb{R}^{m+1} \quad \text{for each moment } t,$$

Where the trivial first entry is set for constant terms. Specifically, the forecast for a predictee $Y_t = (1, y_t, y_{t-\tau}, \dots, y_{t-(m-1)\tau}) =: (Y_t(0), Y_t(1), \dots, Y_t(m))$ is going to be given in a locally linear fashion as

$$\widehat{y_{t+1}} = \sum_{j=0}^m C_t(j) Y_t(j),$$

where $C_t(j)$'s are trained by using the historical points from the library set $\{X_t\}_t$. For each predictee Y_t (with a fixed t), the way of training linear combination coefficients C_t is to fit

$$B = AC$$

with $B_s = w(\|X_s - Y_t\|) \cdot x_{s+1}$ and $A_{sj} = w(\|X_s - Y_t\|) \cdot X_s(j)$, where the weighting kernel is given by

$$w(d) = e^{-(\theta d / \bar{d})}.$$

Here, $\bar{d}$ is the mean distance from Y_t to its $k (= m + 1$ as default) nearest neighbors.

Parameters that have to be determined at first, in practice, are chosen to be the one optimizing the out-of-sample forecast. As mentioned in Sec. 2.2, the parameters m and θ are determined by the largest library when NLA is applied.

S3 Text The setting for the simulations of the food chain model

The occurrence of process errors, of which arrival times are denoted by T_k 's, is a Poisson process (with parameter $\lambda = 0.01$). At an arrival time T_k , for every variable we perturb it by scaling it with a random variable that follows a uniform distribution $U(1-\rho, 1+\rho)$ for some $\rho > 0$. Long-tailed distributions such as Gaussian distribution are not chosen in order to prevent the system from experiencing multiple regime shifts within a short time. White Gaussian noises, $\varepsilon_t \sim N(0, \sigma^2)$ are added as measurement errors onto the model dataset after the simulation based on **Table A** and data normalization, except for the regime indicator N which is not involved in NLA. Note that the magnitude of σ^2 can be considered as the noise level that varied in the sensitivity analysis presented in **Fig 6**.

A community is simulated by the explicit Runge-Kutta method of order 5(4) for 5000 time steps, sampled once every 7 time steps (the sample rate is set arbitrarily). **The change point τ is empirically determined as the moment when the absolute value of change of rate of N reaches its maximum, and we choose the 1000-step time series such that $\tau = 300$ for testing NLA.** To obtain the

1 sampling distribution of change point estimates (**Fig 4**), we simulated 200 replicates of model time
2 series for NLA and CPM analysis. Here, $1/\sigma^2$ can be considered as the signal-to-noise ratio.
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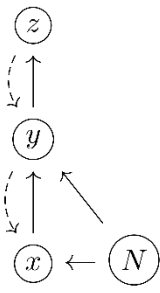
Network	Equations and the parameters
	$\dot{N} = a - bN + \frac{cN^m}{N^m + 1} \quad (b = 0.8, m = 8, c = 1)$ $\dot{x} = x(x - 1) - f_1 y$ $\dot{y} = -d_1 y + f_1 y - f_2 z$ $\dot{z} = -d_2 z + f_2 z \quad (d_2 = 0.01)$ $d_1 = c_0 N + c_1 \quad (c_0 = 0.75, c_1 = 3.7)$ $f_1 = (c_2 N + c_3) \cdot \frac{x}{1 + q_1 x} \quad (c_2 = 0.05, c_3 = 0.4, q_1 = 3.25)$ $f_2 = p_2 \cdot \frac{y}{1 + q_2 y} \quad (p_2 = 0.1, q_2 = 2)$ $a = c_4 t + c_5 \quad (c_4 = 0.375, c_5 = 0.25, 0 \leq t \leq 5000)$

Table A: The settings of the simulations for the pedagogical example in the Section 3.1.

4 **S4 Text NLA analysis for model time series with and without the change point**

5 To present the correctness of NLA analysis in detecting the change point, we generated 400
6 time series replicates of variable y stated in Appendix C that present no clear shifting pattern even
7 under regime shift, among which 200 replicates have no change point and the other 200 have a change
8 point at $t=300$. Both noise strength and library/testing sets are the same with the main results shown in
9 Figure 3. The results were summarized in a confusion matrix presented in **Table A**, where false
10 positive rate and false negative rate are 0.08 (16 out of 200 replicates) and 0 (0 out of 200 replicates),
11 respectively. The problem of minor false positive rate might be further improved by finely controlling
12 the smoothness of RMSE curve using more sophisticated criteria (e.g., generalized cross validation
13 GCV) to avoid misidentification of RMSE valley from false sign changes in the first derivative of
14 RMSE curve.

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		Predicted condition	
		Change point detected	No change point detected
	Total 200 + 200 = 400		
Actual condition	One change point 200	True Positive 200	<i>False Negative</i> 0
	No change point 200	<i>False Positive</i> 16	True Negative 184

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Table A: The confusion matrix of NLA algorithms based on analyzing 400 time series replicates, considering scenarios with and without true change point.

S5 Text Application of NLA on different system variables of the food chain model

In Sec. 3.1, we apply our detection method on the variable y of the food-chain model. Here, we show the results when the same experiment is done using the system variable x and z instead. With **Fig A**, we can again conclude that NLA better detects and estimates the occurrence of regime shift when analyzing the system variable x . When the input is replaced with the time series data of z , CPMs based on Student- t test and Mann-Whitney test give good estimations. Comparatively, NLA is more accurate but less precise. The results of the experiments done with each of system variables are summarized in **Table 1**. The comparison with CPMs suggests that NLA is less dependent of the choice of input variable, and the performance is good in general.

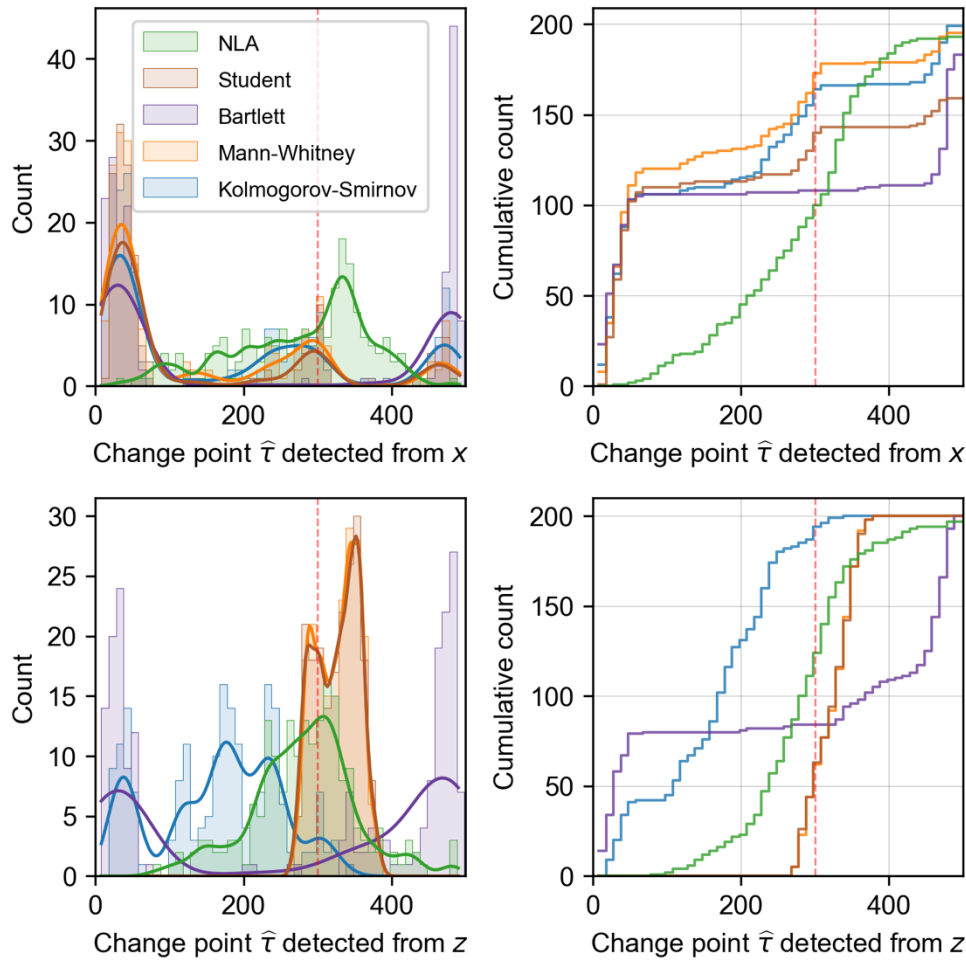


Fig A: The comparison between NLA and CPMs on x and z .

S6 Text Efficacy of NLA in analyzing time series underwent flickering regime shift

We examined the efficacy of NLA method in analyzing the time series manifesting flickering regime shift in which there is no point-wise change point but a rather long transient period between

two regimes (**Fig A**). We analyzed the model time series generated by the flickering model demonstrated in the previous work [1]. The simulation of flickering model repeated in 200 replicates. The upper panel of **Fig A** presents the computed median (blue line), 95%, and 5% quantiles (grey lines) of the time series replicates. Based on the computed median, the flickering dynamics occurred roughly between $t=150$ and $t=350$ and the red dashed line labeled the mid-point of the transient period at $t=250$. When applying NLA on the 200 time series replicates, the detected change points mostly located within the flickering period, indicating the effectiveness of NLA method in revealing the regime shift with long transient dynamics.

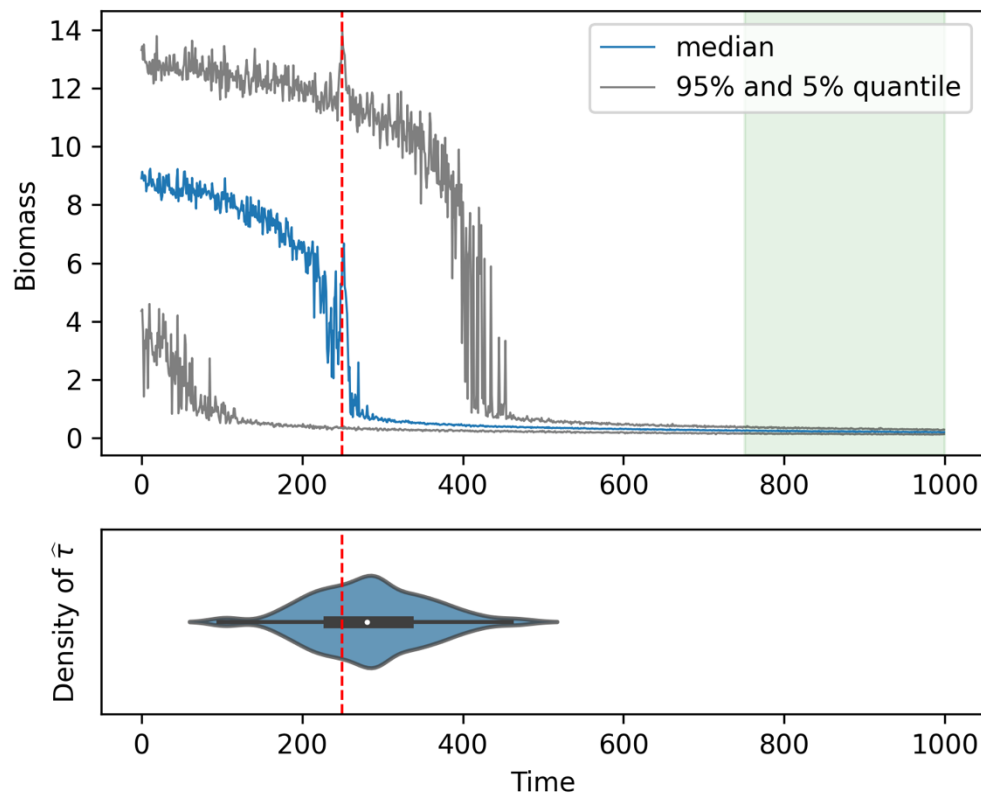


Fig A: NLA applied to the time series underwent flickering regime shift.

S7 Text Efficacy of NLA in sparse time series

We examined the efficacy of NLA method based on a model time series composed of more sparse data points. We reduced the sample size of model time series into a half of that used in original analysis (**Fig 3**) by doubling the sampling intervals. The result based on analyzing the sparse dataset (**Fig A**) indicates that change point can still be correctly detected from the valley of prediction error. However, compared to the findings using full sample size, the shape of prediction valley is shallower than that presented in **Fig 3**, suggesting the signal of change point became weaker when the time series becomes sparse.

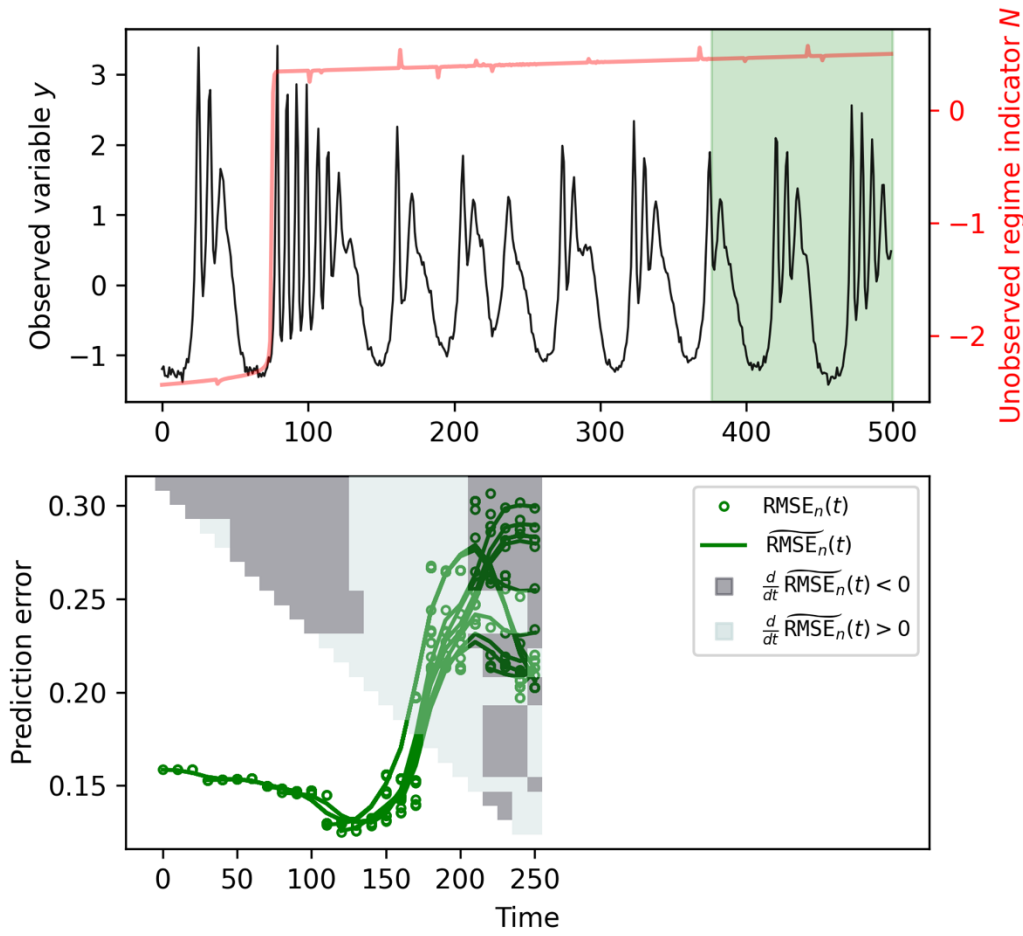


Fig A: NLA applied to the time series of y based on half of the sample size (an analogue of **Fig 3**).

S8 Text Robustness of NLA to the selection of test set located at either ends of the time series.

NLA was applied to the model time series generated by the simulation stated in the Appendix C. To make a fair comparison, we included earlier time points that leads to a symmetric arrangement of the two test sets. We obtained the sampling distribution of the change point estimates by repeating the two NLA analyses on 150 replicates. Our results (**Fig A**) indicate that the sampling distributions

- 1 obtained from forecasting the first quarter (blue) and last quarter (orange) are very close to each other.
- 2 The sampling distribution considering the differences between the two estimates obtained from the
- 3 same replicate also locates around zero, indicating robustness of NLA regardless of test set location.

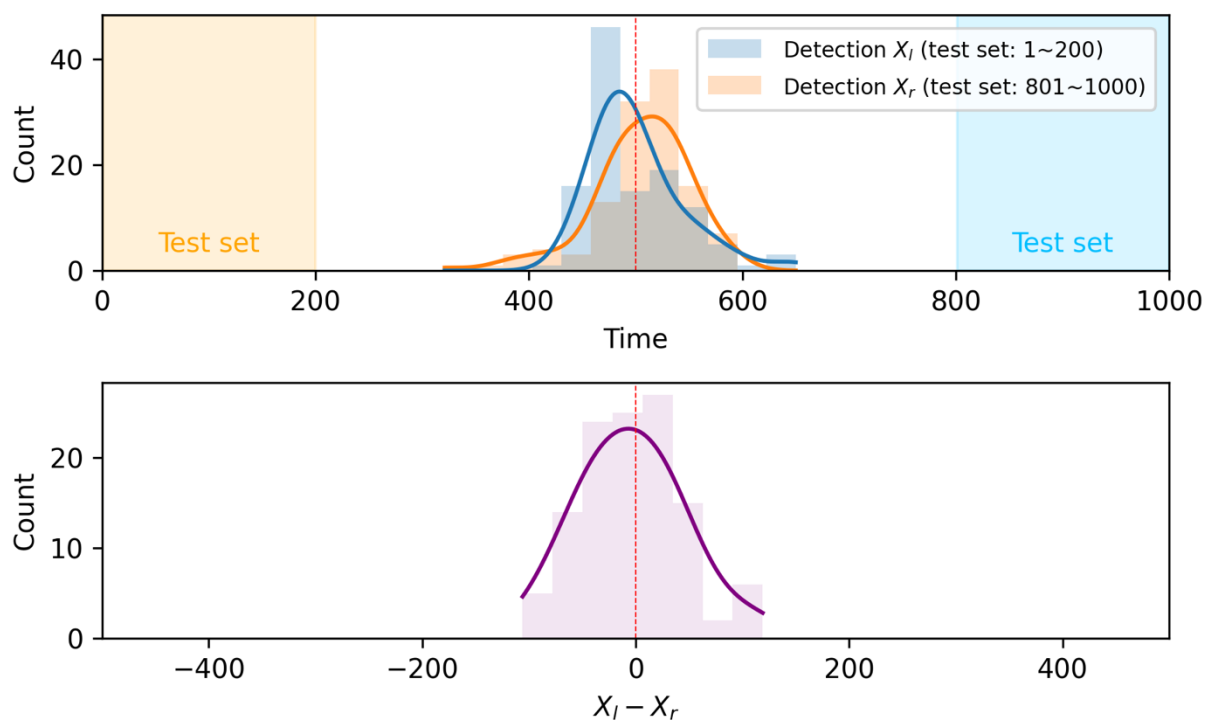


Fig A: Sampling distribution of change point estimates based on NLA analyses that allocate test sets in the first and last quarters.